AN EFFECTIVE CARTIER DIVISOR EXTENDS ACROSS A UFD LOCUS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file divisors.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Carrier window 3: statement and proof both lie inside the window 6000-8000 granted to this lane by the owner lane lane-s13 on 2026-10-14.

Editorial note. The references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: sources/stacks-divisors/retained-preamble.tex (88 lines, verbatim present).

Allowed standard prerequisites (editorial).

- An immersion into a locally Noetherian scheme is quasi-compact. Properties, Lemma properties-lemma-immersion-into-noetherian, used at proof lines 7103-7105. Makes D -> X quasi-compact.
- Formation of the scheme theoretic image commutes with opens for quasi-compact maps. Morphisms, Lemma morphisms-lemma-quasi-compact-scheme-theoretic-image, used at proof lines 7105-7107. Reduces to X affine.
- A divisor is Cartier in a point when its ideal is principal there. Lemma lemma-effective-Cartier-in-points, used at proof lines 7109-7112. Reduces further to the local case.
- Decomposition of an effective Cartier divisor into integral pieces. Lemma lemma-effective-Cartier-divisor-is-a-sum and Lemma lemma-effective-Cartier-codimension-1, used at proof lines 7115-7122. Produces the height one primes and their prime elements.

Source reference. Original lemma 0BCN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14286–14286. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 0BCP: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14287–14287. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 0AG8: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13186–13186. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 081I: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10218–10218. Navigation pointer into the Stacks

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Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 01R8: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2293–2293. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 01OX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2210–2210. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let X be a locally Noetherian scheme. Let $U \subset X$ be an open and let $D \subset U$ be an effective Cartier divisor. If $\mathcal{O}_{X,x}$ is a UFD for all $x \in X \setminus U$, then there exists an effective Cartier divisor $D' \subset X$ with $D = U \cap D'$.*

Proof. Let $D' \subset X$ be the scheme theoretic image of the morphism $D \to X$. Since X is locally Noetherian the morphism $D \to X$ is quasi-compact, see Properties, Lemma 01OX. Hence the formation of D' commutes with passing to opens in X by Morphisms, Lemma 01R8. Thus we may assume $X = \mathrm{Spec}(A)$ is affine. Let $I \subset A$ be the ideal corresponding to D'. Let $\mathfrak{p} \subset A$ be a prime ideal corresponding to a point of $X \setminus U$. To finish the proof it is enough to show that $I_\mathfrak{p}$ is generated by one element, see Lemma 0AG8. Thus we may replace X by $\mathrm{Spec}(A_\mathfrak{p})$, see Morphisms, Lemma 081I. In other words, we may assume that X is the spectrum of a local UFD A. Then all local rings of A are UFD's. It follows that $D = \sum a_i D_i$ with $D_i \subset U$ an integral effective Cartier divisor, see Lemma 0BCP. The generic points ξ_i of D_i correspond to prime ideals $\mathfrak{p}_i \subset A$ of height 1, see Lemma 0BCN. Then $\mathfrak{p}_i = (f_i)$ for some prime element $f_i \in A$ and we conclude that D' is cut out by $\prod f_i^{a_i}$ as desired. $\square$